Name:_____

Exam 2 - MATH-262-01

Instructions: You have 110 minutes to take this exam. Please answer each question in the space provided. Answers without justification may ***earn no credit***. You are permitted your signed prep work journal and a calculator on this exam. No other notes or electronic devices of any kind are allowed on this exam. Failure to comply may result in earning a 0 on any portion of this exam and/or being reported to the university.

1. [20 points] Consider the ODE

$$2t^2u'' + tu' - 3u = 0.$$

(a) Show that $y_1 = \frac{1}{t}$ is a solution on $(0, \infty)$.

(b) Use a reduction of order technique to find a second particular solution.

2. [23 points] Use the method of undetermined coefficients to solve the IVP

$$y'' - 4y' - 12y = 3e^{5x};\ y(0) = \frac{18}{7},\ y'(0) = -\frac{1}{7}.$$

3. [23 points] Use variation of parameters to solve the ODE

$$2y'' + y' - y = x + 1.$$

Hint. Recall $\int e^{ax}(x+1)\,dx = \frac{e^{ax}(ax+a-1)}{a^2} + C.$

4. [32 points] According to ***Kirchoff's Law***, the impressed voltage $E(t)$ on a circuit is the sum of the potential drops across all components. In an LCR circuit, that is

$$V_L + V_R + V_C = E(t).$$

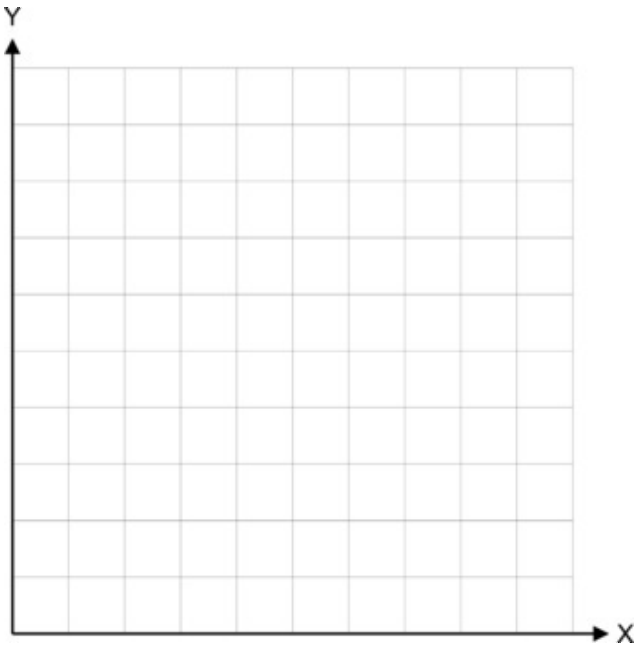
where V_L is the potential drop across the inductor, V_R is the potential drop across the resistor, and V_C is the potential drop across the capacitor. Suppose the circuit has inductance L, resistance R, and capacitance C.

(a) Use the statements below to write a 2nd-order IVP for the charge on the capacitor q at time t.

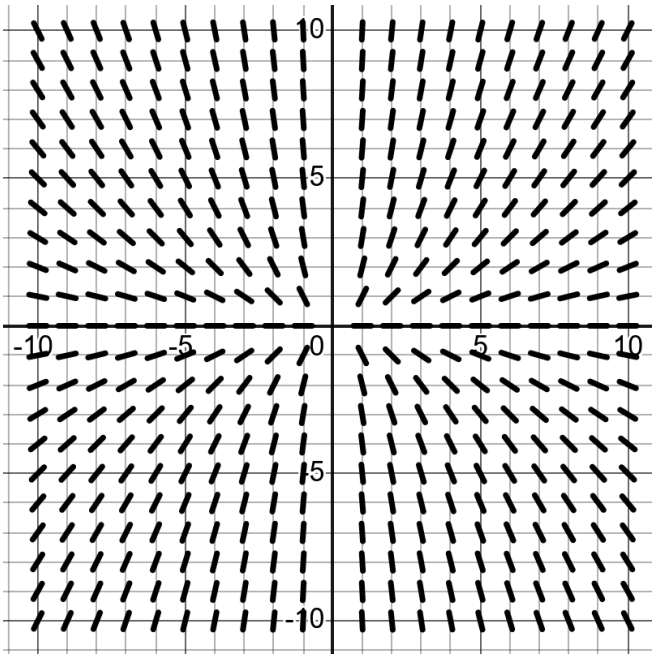
1. The potential drop across a capacitor is the ratio between its charge and capacitance.
2. Current is the rate of change of the charge on the capacitor.
3. The potnetial drop across a resistor is the product of the resistance and the current.
4. The potential drop across an inductor is the product of the inductance and the rate of change of the current.
5. The capacitor starts with no charge or current.

(b) Suppose there is no impressed voltage ($E(t) = 0$ for all t). Use the ODE you built in part (b) to construct an algebraic model for the charge on the capacitor at time t.

(c) Suppose the capacitor must have a capacitance of 10 pF. Sketch a diagram of the (R, L) parameter space below. Label regions of underdamping, critical damping, and overdamping. Specify the region of this parameter space that your solution falls into.



5. [12 points] Suppose the slope field below corresponds to a first-order ODE.



(a) Identify all equilibria (nullclines) for the system. Sketch them onto the picture above, then write their equation(s) in the space below.

(b) Label at least one nonzero isocline for the system. Sketch it onto the picture above, then write their equation(s) in the space below.

(c) Consider the initial condition $y(3) = 1$. Sketch the corresponding trajectory on $(0, \infty)$ onto the picture above.

(d) Consider the initial condition $y(-3) = 1$. Sketch the corresponding trajectory on $(-\infty, 0)$ onto the picture above.

(e) Classify each of the equilibria you found in part (a) as stable, unstable, or neither on $(-\infty, 0)$. Justify your answer.